

DHCP & NACL

DHCP automatically assigns: IP address, Subnet mask, Gateway, DNS to devices on a network.

1. Create DHCP option set

Edit vpc setting → 1. Added dns name 2. DNS Setting As Per Requirement
Enable DNS Resolution

The screenshot shows the 'Create DHCP option set' page in the AWS Management Console. The breadcrumb trail is 'VPC > DHCP option sets > Create DHCP option set'. The page title is 'Create DHCP option set' with an 'Info' link. A descriptive text states: 'Dynamic Host Configuration Protocol (DHCP) provides a standard for passing configuration information to hosts on a TCP/IP network. The options field of a DHCP message contains configuration parameters.'

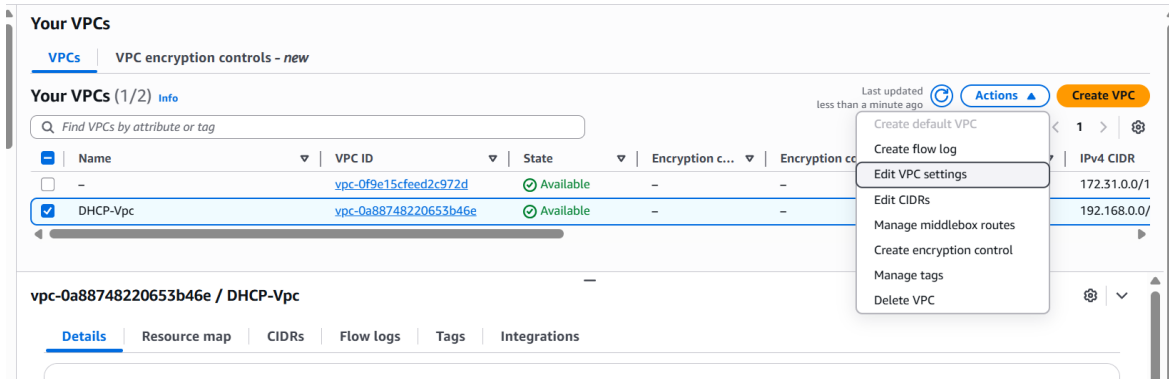
The form is divided into two main sections: 'Tag settings' and 'DHCP option'.
Under 'Tag settings', there is a field for 'DHCP option set name - optional' with the value 'snjb-dhcp'.
Under 'DHCP option', there is a description: 'Specify at least one configuration parameter.'
The 'Domain name' field has the value 'snjb.com'.
The 'Domain name servers' field has the value '10.10.10.1'.
The 'NTP servers' field has the value '172.16.16.16, 10.10.10.10, 75:ff9b:c20, 75:ff9b:c50'.
The 'NetBIOS name servers' field has the value '192.168.0.4, 198.168.0.5'.
The 'NetBIOS node type' field is currently empty.

2. Create VPC

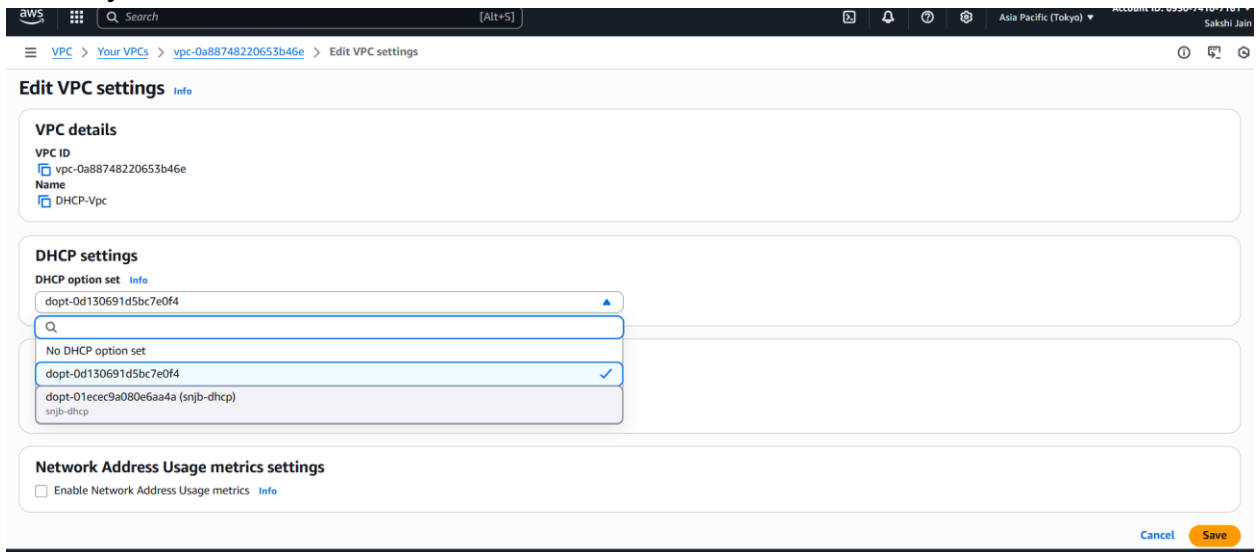
The screenshot shows the 'Create VPC' page in the AWS Management Console. The breadcrumb trail is 'VPC > Your VPCs > Create VPC'. The page title is 'VPC settings' with an 'Info' link. A descriptive text states: 'Create only the VPC resource or the VPC and other networking resources.'

The form is divided into several sections:
Under 'Resources to create', the 'VPC only' radio button is selected.
Under 'Name tag - optional', there is a field for 'Name tag - optional' with the value 'DHCP-Vpc'.
Under 'IPv4 CIDR block', the 'IPV4 CIDR manual input' radio button is selected. The 'IPv4 CIDR' field has the value '192.168.0.0/16'. A note below states: 'CIDR block size must be between /16 and /28.'
Under 'IPv6 CIDR block', the 'No IPv6 CIDR block' radio button is selected.
Under 'Tenancy', the 'Default' dropdown menu is open, showing 'Default' as the selected option.
At the bottom, there is a section for 'VPC encryption control (\$)' with a '- new' link and an 'Info' link. A note states: 'Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. Additional charges apply.'

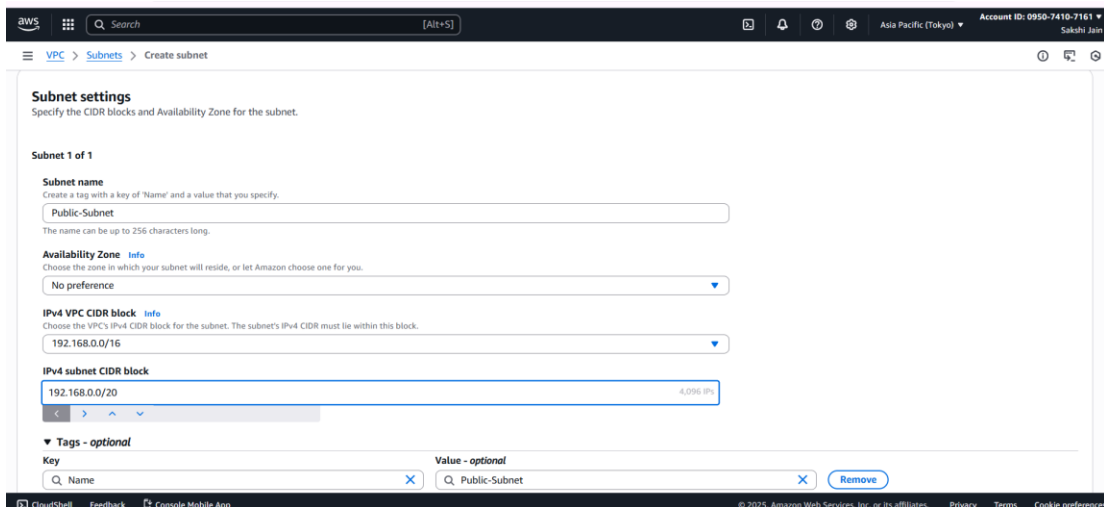
Click on edit VPC settings



Select your DHCP and click on save



3. Create Subnet



You have successfully created 1 subnet: subnet-0c6961a469c20ec1a

Subnets (1/1) Info

Find subnets by attribute or tag

Subnet ID : subnet-0c6961a469c20ec1a X Clear filters

Name	Subnet ID	State	VPC
Public-Subnet	subnet-0c6961a469c20ec1a	Available	vpc-0a88748220653b46e D

subnet-0c6961a469c20ec1a / Public-Subnet

Details Flow logs Route table Network ACL CIDR reservations Sharing Tags

Edit subnet settings Info

Subnet

Subnet ID
subnet-0c6961a469c20ec1a

Name
Public-Subnet

Auto-assign IP settings Info

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address Info

☐ Enable auto-assign customer-owned IPv4 address Info

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings Info

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch Info

☐ Enable resource name DNS AAAA record on launch Info

Hostname type Info

4. Create Internet gateway

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

IGW-1

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q IGW-1 X Remove

Add new tag

You can add 49 more tags.

Cancel Create Internet gateway

Attach to VPC

Attach to VPC (igw-0546b62a231a32375) Info

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

Q vpc-0a88748220653b46e X

AWS Command Line Interface command

Cancel Attach internet gateway

5. In Route Table – Edit Routes and Subnet association

The screenshot displays the AWS Management Console interface for editing a route table. The top section, titled "Route tables (1/2)", shows a list of route tables. The selected route table, "rtb-0ee3c650a8bead9e0", is highlighted. Below this, the "Routes (1)" section shows a single route with destination "192.168.0.0/16", target "local", and status "Active". The "Edit routes" section allows adding new routes, with fields for destination, target, status, propagated, and route origin. The "Edit subnet associations" section shows the available subnets, with "Public-Subnet" selected. The "Selected subnets" section shows the selected subnet, "subnet-0c6961a469c20ec1a".

Route tables (1/2)

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Own
Route-Table	rtb-0ee3c650a8bead9e0	-	-	Yes	vpc-0a88748220653b46e DH...	095
-	rtb-0ff5d70edff3fe16	-	-	Yes	vpc-0f9e15cfeed2c972d	095

rtb-0ee3c650a8bead9e0 / Route-Table

Details | **Routes** | Subnet associations | Edge associations | Route propagation | Tags

Routes (1)

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/16	local	Active	No	Create Route Table

Edit routes

Destination: 192.168.0.0/16 | Target: local | Status: Active | Propagated: No | Route Origin: CreateRouteTable

Q: 0.0.0.0/0 | X | Internet Gateway | - | No | CreateRoute | Remove

Q: igw-0546b62a231a3237f | X

Add route | Cancel | Preview | Save changes

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/1)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Public-Subnet	subnet-0c6961a469c20ec1a	192.168.0.0/20	-	Main (rtb-0ee3c650a8bead9e0 / Route...

Selected subnets

subnet-0c6961a469c20ec1a / Public-Subnet | X

Cancel | Save associations

6. Create EC2 Instance

The screenshot displays the AWS Management Console "Launch an instance" wizard. The "Key pair name" is set to "vpc2". The "Network settings" section shows the VPC as "vpc-0a88748220653b46e (DHCP-Vpc)" and the Subnet as "Public-Subnet". The "Auto-assign public IP" is set to "Enable". The "Firewall (security groups)" section shows the "Create security group" option selected. The "Summary" section shows the number of instances as 1, the software image as "Amazon Linux 2023 9.2", the virtual server type as "t3.micro", the firewall as "New security group", and the storage as "1 volume(s) - 8 GiB". The "Launch instance" button is highlighted.

Key pair name - required

vpc2 | Create new key pair

Network settings

VPC - required

vpc-0a88748220653b46e (DHCP-Vpc) | 192.168.0.0/16

Subnet

subnet-0c6961a469c20ec1a | Public-Subnet

Auto-assign public IP

Enable

Firewall (security groups)

Create security group | Select existing security group

Security group name - required

launch-wizard-1

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 9.2

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Cancel | Launch instance | Preview code

- **Lower rule number = higher priority**
- Rules are evaluated **from smallest number to largest**
- As soon as **one rule matches**, NACL **stops** checking further rules.

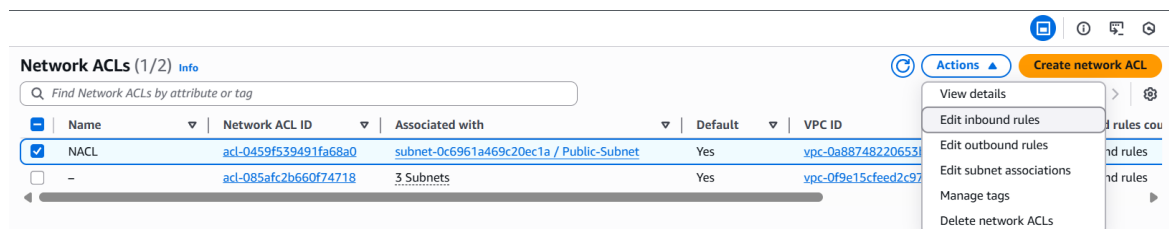
Feature	Security Group	Network ACL (NACL)
Level	Instance level	Subnet level
Type	Stateful	Stateless
Rules	Only allow	Allow + Deny
Default Behavior	All inbound denied, outbound allowed	All inbound/outbound allowed
Use Case	Protect individual EC2 instances	Protect subnets (broader control)

When to Use NACL?

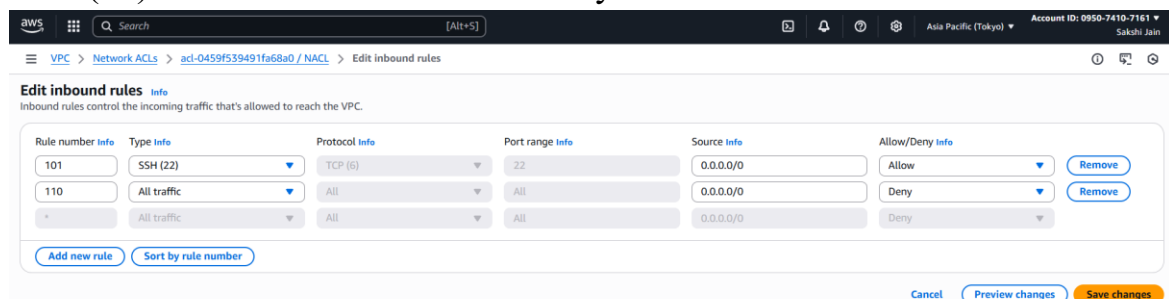
Use NACL when you need:

- Extra layer of security
- Block specific IPs or networks
- Open or close ports for whole subnet

1. Edit Inbound Rule of NACL



SSH (22) – Allow and All traffic – Deny



Check your instance is connected because SSH (22) is having High Priority.

```
aws
Search [Alt+S]
Asia Pacific (Tokyo) Account ID: 0950-7410-7161 Sakshi Jain

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Fri Nov 28 15:04:32 2025 from 3.112.23.5
[ec2-user@ip-192-168-9-93 ~]$
```

Now, SSH (22) – Deny and All traffic – Allow

awsSearch [Alt+S]Asia Pacific (Tokyo)Account ID: 0950-7410-7161Sakshi Jain

VPC > Network ACLs > acl-0459f539491fa68a0 / NACL > Edit inbound rules

Edit inbound rulesInfo

Inbound rules control the incoming traffic that's allowed to reach the VPC.

Rule numberInfo	TypeInfo	ProtocolInfo	Port rangeInfo	SourceInfo	Allow/DenyInfo	
101	SSH (22)	TCP (6)	22	0.0.0.0/0	Deny	Remove
110	All traffic	All	All	0.0.0.0/0	Allow	Remove
*	All traffic	All	All	0.0.0.0/0	Deny	

Add new rule

Sort by rule number

Cancel

Preview changes

Save changes

awsSearch [Alt+S]Asia Pacific (Tokyo)Account ID: 0950-7410-7161Sakshi Jain

Failed to connect to your instance
Error establishing SSH connection to your instance. Try again later.

Check your instance is Failed to connect because SSH (22) is having High Priority and its Deny.